

Department of Computer Science

MSc. AIML Sem1

**Subject: Artificial Intelligence
Theory Assignment – I**

1. Write three definitions of Artificial Intelligence. Write the characteristics of AI.
2. Explain the Turing Test in detail with figures.
3. Write the applications of AI in different areas.
4. Write the algorithm of Strategy – 3 of Tic – Tac – Toe.
5. Explain is Tic – Tac – Toe board extensible to a 4 x 4 for 2 players? Justify.
6. Write about Intelligent Agents in detail. (draw figure)
7. Explain reactiveness and proactiveness of an intelligent agent.
8. Write the Environment types for following (with reason):
 - a. Automatic taxi driver
 - b. AI tutor
 - c. Vacuum cleaner agent
 - d. Traffic control agent
9. Write PEAS description for following:
 - a. Automatic taxi driver
 - b. AI tutor
 - c. Vacuum cleaner agent
 - d. Traffic control agent
10. Explain different types of Agents, in detail.

Some Critical thinking Question for preparation:

1. How would you differentiate between “AI that thinks like humans” and “AI that acts rationally”? Can you give examples where these two approaches might lead to different solutions?

2. If an AI system produces correct results but uses a completely different process from human reasoning, should it still be called “intelligent”? Why or why not?
3. Which task domains do you think are hardest for AI to master and why?
4. When choosing between search-based AI techniques and knowledge-based AI techniques, what factors should guide the choice?
5. Give an example of an environment that is dynamic, stochastic, and multi-agent. How would these properties influence the agent’s architecture?
6. Why is it important to formally classify environments (e.g., deterministic vs. stochastic) before designing an AI agent?
7. Can an agent be considered rational even if its decisions sometimes lead to failure? Under what conditions?
8. What are the main challenges in keeping a knowledge base consistent as an AI learns from new data?
9. How would you redesign a simple reflex agent to perform better in an environment that changes unpredictably?
10. What are the trade-offs between a goal-based agent and a utility-based agent in a real-world application like autonomous driving?